Shivanka Priyashan

Software Engineer

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LinkedIn | HackerRank | Reddit

PROFILE

I am a highly motivated individual with a strong passion for solving problems. I thrive in collaborative environments and constantly seek opportunities to expand my knowledge and skills. I am dedicated to pursuing a career in Computer Vision, Machine Learning, Deep Learning, and NLP where I can leverage my skills and passion to make meaningful contributions.

INTERESTS

Machine Learning | **Deep Learning** | **Computer Vision** | **Natural Language Processing**

WORK EXPERIENCE

Software Engineer, *Global Wavenet, Sri Lanka* — Sep 2024 – present

- Development of AI-driven solutions using **Azure** and **OpenAI** services
- Contributed to the development of an immersive marketplace with virtual assistant on **Meta Quest 2** and **Apple Vision Pro**

Software Engineer Intern, *Zebra Technologies, Sri Lanka* — Jan 2023 – Jul 2023

Assisting EMC Data Capture team in extending portfolio in Machine Learning/Computer Vision domain

- **Project 1**: Developed method to detect and count number of barcodes present in a region [*Python | OpenCV*]
- **Project 2**: Developed method to perform OCR on custom font [*Python | OpenCV*]

Tutor — 2019 – present

Private tutoring for Advanced Level students in Combined Mathematics

EDUCATION

B.Sc. (Hons.) Electronic and Telecommunication Engineering, — Jan 2020 – Jun 2024
University of Moratuwa, Sri Lanka

CGPA - **3.79 / 4.2** | Deans' List : Semester **1, 2, 6, 7, 8**
Machine Vision **(A+)** | Robotics **(A+)** | Autonomous Systems **(A)**
Fundamentals of Image Processing and Machine Vision **(A)**
Data Structures and Algorithms **(A)** | Linear Algebra **(A)**
Embedded Systems Engineering **(A+)**

G.C.E Advanced Level, *Loyola College, Negombo, Sri Lanka* — 2018

Combined Mathematics **(A)**, Physics **(A)**, Chemistry **(A)**, General English **(A)**
Z score - **1.9515**

COURSES

Deep Learning Specialization

Online course offered by **DeepLearning.AI** through **Coursera** (ongoing)
Course 1 - Neural Networks and Deep Learning **[May 2023]**

Course 2 - Improving Deep Neural Networks **[October 2023]**
Course 3 - Structuring Machine Learning Projects **[June 2024]**
Course 4 - Convolutional Neural Networks (Auditing)
Course 5 - Sequence Models (Auditing)

Machine Learning 2022
Online course offered by the **University of Stanford** through **Coursera**

First Principles of Computer Vision Specialization
Online course offered by **Columbia University** through **Coursera** (Auditing)
Course 1 - Camera and Imaging
Course 2 - Features and Boundaries
Course 3 - 3D Reconstruction - Single Viewpoint
Course 4 - 3D Reconstruction - Multiple Viewpoints
Course 5 - Visual Perception

Parallel and Concurrent Programming with C++ Part 1 2024
Online course offered through **LinkedIn Learning**

C++ Best Practices for Developers 2024
Online course offered through **LinkedIn Learning**

C++ Templates and the STL 2023
Online course offered through **LinkedIn Learning**

Python Data Structures 2022
Online course offered by the **University of Michigan** through **Coursera**

{ } SKILLS

Languages
Python | C++ | MATLAB

Softwares
SOLIDWORKS | Altium | Webots | Coppelia Sim

Libraries and Frameworks
Tensorflow | Pytorch | Keras | OpenCV | Numpy | ROS | Pandas

Microcontrollers
ATmega328P | ESP32 | Raspberry Pi

PROJECT EXPERIENCE

[Final Year Project] Predictive Maintenance with AI-Powered Digital Twins, Sep 2023 – Apr 2024
Python | C++ | Pytorch | Altium | ESP32 | Microsoft Azure
Utilizing Industrial Internet of Things (IIoT) based Digital Twin models, real-time sensor data and machine learning algorithms, to detect faults, predict root causes, and forecast the remaining useful life of a vertical axis wind turbine and battery system in a micro-grid

Football Analysis using Computer Vision, *Python | OpenCV | YOLO* 2024
Using OpenCV and YOLO to analyze football matches which includes tasks such as player tracking, and ball tracking.

Automatic Number Plate Recognition, *Python | OpenCV | YOLO | EasyOCR* 2024
Automatic Number Plate Recognition (ANPR) system using YOLOv8 for object detection and EasyOCR for text recognition.

Mecanum Wheel Robot Car, *Python | RaspberryPi | ROS (ongoing)* 2023
Mecanum wheel robot car project using RaspberryPi 3B+ with ROS. Includes four DC motors controlled using two L298N motor drivers

Robot Design and Competition, *C++ | SOLIDWORKS | OpenCV | Webots*
Successfully designed virtual robot that performs solid line following, dotted line following, maze traversal, object detection, object pick and place.

LPG Gas Leakage Detector and Indicator, *C++ | Solidworks | Altium | ESP32*
Designing and implementing a LPG gas leakage detector and indicator which notifies a user about a leakage through an IoT framework.

Simple Voice Recorder, *C++ | SOLIDWORKS | Altium | Proteus | AVR*
Successfully designed and implemented a simple voice recorder capable of recording, saving, selectively playing back multiple voice recordings and adding adjustments to the spectrum during playback. Project was completed with a validated PCB and enclosure design

AWARDS AND COMPETITIONS

Winner of "AI Innovation Challenge", *Global Wavenet, Sri Lanka* Dec 2024
Awarded in recognition of outstanding innovation and contribution in designing an architectural concept to advance AI solutions based on **Microsoft AI Design Wins**.

National Winner - Best Innovative Product or Project - University Jun 2024
Western Province Winner - Best Innovative Product or Project - University, *Sri Lanka Association for Software and Services Companies (SLASSCOM) - INGENUITY AWARDS 2024*
Won 2 awards for our final year project an **End-to-End Framework for Predictive Maintenance with AI-Powered Digital Twins**

Ranked 4th out of 260 teams at eYantra Robotics Competition, Apr 2022
Python | CoppeliaSim | SOLIDWORKS | OpenCV | Lua
Our theme was set in a simulated environment where indoor vertical farming is explored. Our team was tasked to design, simulate and code our own robotic mechanism that can navigate through different rooms and galleries of the farm and harvest berries based on their size, ripeness and quantity required. Knowledge related to Robotics simulation, Image Processing, 3D designing, Python programming were used.

IDEA Travel Grant, CASE 2023 Aug 2023
Inclusion, Diversity, Equity and Accessibility (**IDEA**) travel awards to participate in Conference for Automation Science and Engineering (**CASE**) 2023 in Auckland, New Zealand offered by IEEE Robotics & Automation Society - Women in Engineering (WiE)

Certificate of Participation - IEEE Extreme 17.0 Programming Competition Oct 2023
Global rank **913** | Country rank **76**

VOLUNTEERING EXPERIENCE

President, *Special Interest Group in Embedded Hardware* Oct 2023 – present

Head of Project Management, *SPARK branch, Electronic Club* 2023 – present

Workshop facilitator, Sri Lanka Challenge Sphere 2024 Jun 2024
Conducted an Arduino workshop for the **Sri Lanka Challenge Sphere 2024, Arduino Challenge.**

Department Coordinator, *EXMO 2023* 2023
Responsible for organizing and managing lab tours, bootcamps and competitions, managing budget for EXMO 2023 at the department of Electronic and Telecommunication Engineering, University of Moratuwa

Group Leader, Sri Lanka Robotics Challenge 2022 (SLRC) 2022

Responsible for handling several groups of students to share knowledge on work carried at ENTC department and help complete their robots designed for the workshop

Visiting Instructor, Robot Design and Competition (EN2533) 2022

Responsible for conducting practicals of EN2533 Robot Design and Competition module with a group of members. Contributed to check the equipment used in the practicals and complete the practical sheets.

ARTICLES

Combating COVID-19: The Role of Robotics Jan 2022

Article for E-CARRIER, the official magazine of Electronic and Telecommunication Engineering Department, University of Moratuwa

REFERENCES

Dr. Subodha Charles, *Ph.D. (University of Florida, USA), B.Sc. Eng. Hons. (Moratuwa),*
Senior Lecturer,
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